**1** **(a)** $2^2 \times 3 \times 7$ **[B2]**

M1 for $2, 2, 3, 7$ correctly identified in a factor tree or division ladder

**(b)** 252 **[B2]**

M1 for writing $126 = 2 \times 3^2 \times 7$ or writing multiples of 84 and 126

*[Total Marks: 4]*

**2** **(a)** **(i)** 12 or 36 **[B1]**

**(ii)** 64 **[B1]**

**(iii)** 23 **[B1]**

**(b)** Prime factors: $36 = 2^2 \times 3^2$ and $64 = 2^6$ **[M1]**

LCM $= 2^6 \times 3^2 = 576$ **[A1]**

*[Total Marks: 5]*

**3** **(a)** **(i)** 2 or 23 **[B1]**

**(ii)** 27 or 64 **[B1]**

**(iii)** 64 **[B1]**

**(iv)** 15 **[B1]**

**(b)** $2^3 \times 3^2$ **[B2]**

If 0 scored, M1 for $2 \times 2 \times 2 \times 3 \times 3$ or a correct factor tree/division **[M1]**

*[Total Marks: 6]*

**4** **(a)** Irrational **[B1]**

**(b)** Any valid example, e.g. $\sqrt{2}$, $\sqrt{3}$, $\pi$ (if $y^2 = \pi^2$ which is irrational, so this is wrong)

Only square roots of non-square rational numbers, e.g. $\sqrt{2}$ **[B1]**

**(c)** **(i)** 53 and Yes **[B2]**

M1 for substituting $n = 3$: $3^2 + 3 + 41$

**(ii)** Substituting $n = 41$ yields $41^2 + 41 + 41$ **[M1]**

This factorises to $41(41 + 1 + 1) = 41 \times 43$, which has factors other than 1 and itself, therefore it is not prime. **[A1]**

*[Total Marks: 6]*

**5** **(a)** **(i)** $pq$ **[B1]**

**(ii)** $p^3q^3$ **[B1]**

**(b)** $pq = 15$, so the prime numbers are 3 and 5 **[M1]**

$m > n \implies p^3q > pq^3 \implies p^2 > q^2 \implies p > q$ **[M1]**

$p = 5$ and $q = 3$ **[A1]**

**(c)** $m \times n = (p^3q)(pq^3) = p^4q^4$ **[M1]**

$= (p^2q^2)^2$ which is the square of an integer, hence a square number **[A1]**

**(d)** Substitutes values: $m = 5^3 \times 3^1$ **[M1]**

Recognises the power of 3 needs to be a multiple of 3, so $k = 3^2 = 9$ **[A1]**

*[Total Marks: 9]*

**6** $x = 2$ **[B2]**

$y = 3$ **[B2]**

If 0 scored, award M1 for expressing $180 = 2^2 \times 3^2 \times 5$ or $12600 = 2^3 \times 3^2 \times 5^2 \times 7$

*[Total Marks: 4]*

**7 (a)** Identifies HCF as $2^1 \times 3^1$ **[M1]**

$x = 1$ **[A1]**

**(b)** Prime factorises $8100 = 81 \times 100 = 3^4 \times 2^2 \times 5^2$ **[M1]**

Compares highest powers of 3 between $A$ ($3^3$) and $C$ ($3^y$) to the LCM ($3^4$) **[M1]**

$y = 4$ **[A1]**

**(c)** $\sqrt{11}$ **[B1]**

$4\pi$ **[B1]**

*[Total Marks: 7]*

**8 (a)** 12 **[B2]**

If 0 scored, M1 for $84 = 2^2 \times 3 \times 7$ and $120 = 2^3 \times 3 \times 5$ OR for a common factor of 4 or 6 identified **[M1]**

**(b)** 840 **[B2]**

If 0 scored, M1 for $2^3 \times 3 \times 5 \times 7$ **[M1]**

**(c)** $\frac{5}{4}$ or $1\frac{1}{4}$ **[B2]**

If 0 scored, M1 for $\frac{1}{0.8}$ or $\frac{10}{8}$ seen **[M1]**

**(d)** Irrational, because 120 is not a perfect square (or $\sqrt{120}$ cannot be written as a simple fraction) **[B2]**

If 0 scored, B1 for Irrational with no/incorrect reason **[B1]**

*[Total Marks: 8]*

**9 (a)** $x = 2$ **[B1]**

$y = 2$ **[B1]**

M1 for $84 = 2^2 \times 3 \times 7$ OR $3528 = 2^3 \times 3^2 \times 7^2$ seen **[M1]**

**(b)** 1764 **[B1]**

**(c)** 10:21:00 **[B3]**

M2 for LCM of 45 and 72 = 360 seconds (or 6 minutes) **[M2]**

If M2 not scored, M1 for listing multiples of 45 and 72 with at least one common multiple, OR $45 = 3^2 \times 5$ and $72 = 2^3 \times 3^2$ **[M1]**

*[Total Marks: 7]*

**10 (a) (i)** 23 **[B1]**

**(ii)** 8 or 27 or 64 **[B1]**

**(iii)** 64 **[B1]**

**(b)** $2^2 \times 3^2$ (or $2 \times 2 \times 3 \times 3$) **[B2]**

Partial working: Correct factor tree or repeated division showing 2 and 3 as prime factors **[M1]**

*[Total Marks: 5]*

**11 (a)** 36 (or $2^2 \times 3^2$) **[B2]**

Partial working: Identifies $2^2$ and $3^2$ as common prime factors **[M1]**

**(b)** $2^3 \times 3^4 \times 5 \times 7$ **[B2]**

Partial working: One incorrect power or missing prime factor **[B1]**

**(c)** 66 **[B3]**

Partial working: Identifies that $C$ must contain the factor 11 from the LCM **[M1]**

Partial working: Identifies that $C$ must contain exactly $2^1 \times 3^1$ to satisfy the HCF condition **[M1]**

*[Total Marks: 7]*

**12 (a) (i)** $x = 2$, $y = 8$, $z = 10$ **[B2]**

Partial working: Lists correctly prime numbers $x < 10$ and cube numbers $y > 1$ **[M1]**

**(ii)** $x = 7$, $y = 8$, $z = 15$ OR $x = 2$, $y = 64$, $z = 66$ **[B2]**

Partial working: Identifies another valid combination of a prime and cube summing to a triangle number **[M1]**

**(b) (i)** e.g. $p = \sqrt{2}$, $q = -\sqrt{2}$ OR $p = 1 + \sqrt{3}$, $q = 1 - \sqrt{3}$ **[B1]**

**(ii)** e.g. $p = \sqrt{2}$, $q = \sqrt{8}$ OR $p = \sqrt{3}$, $q = 2\sqrt{3}$ **[B1]**

*[Total Marks: 6]*

**13** **(a)** 13 **[B1]**

**(b)** $\sqrt{7}$ **[B1]**

**(c)** $-5$ **[B1]**

**(d)** 0.25 **[B1]**

**(e)** 64 **[B1]**

*[Total Marks: 5]*

**14** **(a)** 36 (or $2^2 \times 3^2$) **[B1]**

**(b)** $2^3 \times 3^3 \times 5 \times 7$ **[B2]**

If 0 scored, allow [M1] for a product with at least three correct prime bases and powers, or for the correct evaluation 7560.

**(c)** 10 **[B2]**

If 0 scored, allow [M1] for identifying required missing powers $2 \times 5$ or $2^1 \times 5^1$.

*[Total Marks: 5]*

**15** **(a)** 40 **[B2]**

If 0 scored, allow [M1] for setting up the equation $12x = 4 \times 120$ or finding prime factors of 120 indicating $2^3 \times 5$.

**(b)** **(i)** $2^4 \times 3^2 \times 7$ **[B2]**

If 0 scored, allow [M1] for a correct factor tree or continuous division with at least 2 correct prime division steps.

**(ii)** 216 **[B3]**

If 0 scored, allow [M1] for writing $72 = 2^3 \times 3^2$.

Allow a further [M1] for deducing that $N$ must contain $2^3 \times 3^3$ to be the smallest perfect cube with the correct HCF (power of 2 is exactly 3, power of 3 is at least 3).

*[Total Marks: 7]*